

Notes on grifting by optimal transport

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Consider an affine measurable space $\mathcal{S} \subset \mathbb{R}^d$. Fixing $(\Omega, \Sigma, \mathbb{P})$, $\Omega = \mathcal{S}$, $\Sigma = \mathcal{B}(\mathcal{S})$ the Borell σ -algebra of $\mathcal{S}$, and $\mathbb{P}$ a probability measure on (Ω, Σ) . Let $v := (v_t)_{t \in \mathbb{N}}$ be a collection of Σ -measurable automorphisms on $\mathcal{S}$. For any random variable $X : \Omega \rightarrow \mathcal{S}$, let $\mathcal{L}(X)$ denote the law of X under $\mathbb{P}$.

The dynamical system of a drifting model (Deng et al., 2026) evolves the prediction X during training according to the following Markov chain:

$$X_{t+1}^v = X_t^v + v_t(X_t) \in \mathcal{S} \quad (1)$$

where $v : (t, x) \in \mathbb{N} \times \mathcal{S} \rightarrow \mathcal{V} \subset \mathcal{S}$. Equivalently, letting μ_0 be a probability measure on (Ω, Σ) , this can be viewed in terms of the evolution of the law of X_t from its initial distribution μ_0 as

$$\begin{cases} \mathcal{L}(X_{t+1}^v) = (\text{Id} + v_t)_\# \mathcal{L}(X_t^v) \\ \mathcal{L}(X_0^v) = \mu_0 \end{cases} \quad (2)$$

The development of drifting models Deng et al. (2026) rests on the idea of crafting (by hand) a vector-field-valued map v which is asymptotically stable and whose attracting equilibrium induces $X_\infty \sim \nu$ for a given target distribution $\nu \in \mathcal{P}(\mathcal{S})$. By using the methodology of least-squares fitting outlined in Deng et al. (2026), the specification of v can be leveraged into the *training* of a neural network f which yields an approximate pushforward map transforming a given noise distribution μ_0 into the target distribution ν , i.e. $\nu \approx f_\# \mu_0$.

From a technical standpoint, the authors of Deng et al. (2026) construct a family of vector fields v depending on ν and $\mathcal{L}(X_t^v)$ which admits as an equilibrium the target distribution ν . However, under general conditions it remains unclear whether the resulting mean-field dynamics of X_t will meaningfully converge to ν as $t \rightarrow \infty$, despite them giving some sufficient conditions.

A contrario, it is tempting to reinterpret (1) as a control problem, and try and determine v through the specification of an objective function to be optimised. At an intuitive level, we can then view the efforts of Deng et al. (2026) as an ad-hoc attempt to guess a control v which will yield the desired target distribution ν .

Inspired by the recent work of Müller et al. (2026), we propose to study this as a *measure constrained* optimal control problem.

1. Proposed setting

Let $\gamma \in [0, 1)$, $c : \mathcal{S}^2 \rightarrow \mathbb{R}$ be a measurable cost function, and let $\mu_0 \in \mathcal{P}(\mathcal{S})$ be a given initial distribution, and consider the optimal control problem,

$$\begin{aligned} J_p : \mu_0 \in \mathcal{P}(\mathcal{S}) \mapsto \min_{v \in \mathcal{V}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, X_{t+1}^v) \right] \\ \text{subject to } X_{t+1}^v = X_t^v + v_t(X_t^v) \text{ for all } t \geq 0, \\ X_0^v \sim \mu_0, \end{aligned} \quad (3)$$

where $\mathcal{V}$ is a set of admissible controls. By abuse of notation, we will write $J_p(x)$ for the value of the optimal control problem when $\mu_0 = \delta_x$ is a Dirac measure at $x \in \mathcal{S}$. It is easy to see that a dynamic programming principle holds for J_p , namely

$$J_p(x) = \min_{\bar{v} \in \mathcal{S}} \{c(x, x + \bar{v}) + \gamma J_p(x + \bar{v})\}. \quad (4)$$

and that thus

$$J_p(\mu_0) = \int J_p(x) \mu_0(dx). \quad (5)$$

The optimal policy $v^* \equiv v^*$ is then readily extracted from the minimiser of the right-hand side of the dynamic programming principle.

Consequently, for any x_0 , optimal trajectories of the system (1) follow the constant speed curves

$$t \mapsto \sum_{s=0}^{t-1} v^*(x_s) = (\text{Id} + v^*)^t x_0, \quad (6)$$

where $x_{s+1} = x_s + v^*(x_s)$. This suggests an alternative fitting approach than the one proposed in [Deng et al. \(2026\)](#): find an approximate fixed point of J_p using least-squares and a Neural Network, somehow extract v^* (for example with c being a squared norm to obtain the network gradient), and then predictions can be computed by feeding forward through the network a few times. However the issue is we don't have a good way to ensure $X_t \rightarrow \nu$ from this. But it could give us ideas for after we include the constraint $\mathcal{L}(X_t^v) \rightarrow \nu$ in the optimal control problem.

To reduce the simulation cost, we can also try to find an efficient representation of $(\text{Id} + v^*)^t$ for t either fixed or as an input parameter of a neural network approximating it (this gets us closer in spirit to the simulation-free approach of drifting models). If v^* is a gradient field, then we can use the fact that $(\text{Id} + v^*)^t$ is the gradient of a convex potential field perhaps, by writing this as a gradient flow (this would mirror the approach of [Müller et al. \(2026\)](#)). If v^* is itself a linear operator, then we can directly operate on the spectrum of v^* to compute the exponential for any t very easily. How to justify that v^* is a linear operator is unclear, maybe we can use a latent space / NTK representation or restrict the control class to linear operators. An approximate alternative is presented by linearising v^* around x_0 and then computing the exponential of the linearised operator. This corresponds to doing a single shot straight path simulation. I suppose we could also try to be clever by combining different points like in a Runge-Kutta scheme to get very good approximations on the cheap.

1.1. Formulations of the simulation constraint

The dynamical system (1), in the equilibrium condition specified by [Deng et al. \(2026\)](#), would ideally satisfy

$$\lim_{t \rightarrow \infty} \mathcal{L}(X_t^v) = \nu \text{ i.e. } \mathcal{L}(X_{t+1}^v) \approx \mathcal{L}(X_t^v) \approx \nu, \quad (7)$$

(understood as in distribution) for t sufficiently large, where $\nu \in \mathcal{P}(\mathcal{S})$ is a given target distribution, and equality is in distribution. If we restrict ourselves to distributions with finite variance (sledgehammer argument: images are bounded), by the metrisation of weak convergence by the Wasserstein-2 distance, this is equivalent to

$$\mathcal{W}_2(\mathcal{L}(X_t^v), \nu) \approx \mathcal{W}_2(\mathcal{L}(X_{t+1}^v), \mathcal{L}(X_t^v)) \approx 0. \quad (8)$$

One path: fix as a limit constraint, combine with the other constraint and relax? The stationarity condition can be equivalently rewritten in terms of the velocity field v_t as

$$\sup_{\varphi \in C_b(\mathcal{S})} \int \varphi(x + v_t(x)) - \varphi(x) \mathcal{L}(X_t^v)(dx) = 0, \quad (9)$$

which can be computed using a GAN but this sup is very bad. Anticipating a bit, if $\mu_t^v = \mathcal{L}(X_t^v, X_{t+1}^v)$ and denoting by $\oplus$ the direct sum, this is a convex constraint on μ_t^v :

$$\mu_t^v \in \mathcal{P}_\gamma(\mathcal{S}^2) : \sup_{\varphi \in C_b(\mathcal{S})} \int \varphi \oplus \varphi(x, y) \mu_t^v(dx, dy) = 0. \quad (10)$$

Equivalently the distributional constraint can be rewritten as

$$\sup_{\varphi \in C_b(\mathcal{S})} \int \varphi(x) \nu(dx) - \varphi(x) \mathcal{L}(X_t^v)(dx) = 0. \quad (11)$$

One remark about imposing stationarity, if v is constant then $\mu := \mathcal{L}(X_t^v) = \mathcal{L}(X_{t+1}^v)$ implies that $\mu = (\text{Id} + v)|_\# \mu$, and thus $\mu \in \ker(\text{Id} + (\text{Id} + v)|_\#)$, understood as a linear operator on signed measures. Note that this is a linear constraint on μ but it is coupled to the primal adjoint variable v .

Another way to impose stationarity is to require that $T := \text{Id} + v$ is admissible as a Monge map for the transport problem between μ and itself, i.e. $T|_\# \mu = \mu$. This can perhaps be relaxed to a Kolmogorov coupling constraint on the joint. Come to think of it, this is exactly a rewriting of the constraints that the joint be the joint of the stationary Markov chain with transition T .

1.2. Dual control

While the primal theory tackles (3) head-on, we will instead focus on the (control-theoretic) dual formulation of the problem, which is linear in the space of measures.

To do so, let $\mathcal{M}_+(\mathcal{S}^2)$ be the space of σ -finite positive measures on $\mathcal{S}^2$, and let $\mathcal{P}(\mathcal{S}^2) \subset \mathcal{M}(\mathcal{S}^2)$ be the space of probability measures on $\mathcal{S}^2$. We define the “forward” and “backward” marginalisation operators $\mathcal{F}$ and $\mathcal{B}$ on $\mathcal{M}_+(\mathcal{S}^2)$ as follows:

$$\mathcal{F} : \mu \in \mathcal{M}_+(\mathcal{S}^2) \rightarrow \int_{\mathcal{S}} \mu(dx, \cdot) \in \mathcal{M}_+(\mathcal{S}) \quad (12)$$

$$\mathcal{B} : \mu \in \mathcal{M}_+(\mathcal{S}^2) \rightarrow \int_{\mathcal{S}} \mu(\cdot, dy) \in \mathcal{M}_+(\mathcal{S}) \quad (13)$$

For any $\gamma \in [0, 1)$, let $\mathcal{P}_\gamma(\mathcal{S}^2) := (1 - \gamma)^{-1} \mathcal{P}(\mathcal{S}^2)$ be the subspace of $\mathcal{M}_+(\mathcal{S}^2)$ of measures with total mass $(1 - \gamma)^{-1}$.

For any $v \in \mathcal{V}$, with μ_0 implicitly fixed, let $\mu^v \in \mathcal{P}_\gamma(\mathcal{S}^2)$ be the measure defined by

$$\mu^v(dx, dy) := \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_t^v, X_{t+1}^v)(dx, dy). \quad (14)$$

This is equivalently written as

$$\mu^v : (A, B) \in \Sigma^2 \rightarrow \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(X_t^v \in A, X_{t+1}^v \in B) = \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(X_t^v \in A, (\text{Id} + v_t)(X_t^v) \in B). \quad (15)$$

The *control-theoretic dual* is the linear problem

$$J_d := \min_{v \in \mathcal{V}} \int c(x, y) \mu^v(dx, dy) \quad (16)$$

$$\begin{aligned} \text{subject to } \mu^v &\in \mathcal{P}_\gamma(\mathcal{S}^2), \\ \mathcal{B}\mu^v &= \mu_0 + \gamma \mathcal{F}\mu^v \end{aligned} \quad (17)$$

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \bigcirc_{s=1}^t (\text{Id} + v_s)|_\# \mu_0$$

Lemma 1 *The control-theoretic duality is strong, i.e., $J_p = J_d$. (We will treat minimiser equivalence later.)*

Proof That $\int c d\mu^v = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, X_{t+1}^v)]$ is immediate from the definition of μ^v .

To establish the constraints of the dual, note that, by definition $\mathcal{B}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_t^v)$, and thus

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_{t+1}^v) = \frac{1}{\gamma} (\mathcal{B}\mu^v - \mu_0). \quad (18)$$

This establishes the constraint associated to $X_0 \sim \mu_0$. The constraint for the dynamics directly leads to

$$\mathcal{F}\mu^v = \sum_{t=0}^{\infty} \gamma^t \mathcal{L}(X_{t+1}^v) = \sum_{t=0}^{\infty} \gamma^t (\text{Id} + v_t)_\# \mathcal{L}(X_t^v) = \sum_{t=0}^{\infty} \gamma^t \bigcirc_{s=1}^t (\text{Id} + v_s)_\# \mu_0. \quad (19)$$

■

1.3. A Remark on minimal energy configurations

If we set $c(x, y) = \|x - y\|^2$, then the optimal control problem (3) becomes

$$\begin{aligned} \min_{v \in \mathcal{V}} \quad & \sum_{t=0}^{\infty} \gamma^t \|v_t\|_2^2 \\ \text{subject to } & X_{t+1}^v = X_t^v + v_t(X_t) \text{ for all } t \geq 0, \\ & X_0 \sim \mu_0, \end{aligned} \quad (20)$$

The constraints vanish in this case, as the objective is independent of the distribution of X_t^v . The optimal control problem then reduces to a standard optimal control problem, and the optimal control is given by $v_t^* \equiv 0$ for all $t \geq 0$.

For this to be useful for the transport problem, we need to balance with the constraints so that the movement is uniform in time rather than concentrated at infinity.

2. Forgetting about discounting

If we set the problem as finite horizon, it is much easier to lay out the constraint we care about for generation.

In particular, we can consider the following finite-horizon optimal control problem, in continuous time, with a terminal constraint:

$$\begin{aligned} J_p(0, \mu_0) &:= \min_{v \in \mathcal{V}_0} \int_0^1 \int c(X_t^v, v_t(X_t^v)) dt d\mu_0 \\ \text{subject to } & \dot{X}_t^v = v_t(X_t^v) \text{ for all } t \in [0, 1], \\ & X_0^v \sim \mu_0, \\ & \mathcal{L}(X_1^v) = \nu \end{aligned} \quad (21)$$

We can then define the cost-to-go functions as $J_p(t, \mu) := \min_{v \in \mathcal{V}_t} \int_t^1 c(X_s^v, v_s(X_s^v)) ds$ subject to $\dot{X}_s^v = v_s(X_s^v)$ for all $s \in [t, 1]$, $X_t^v \sim \mu$, and $\mathcal{L}(X_1^v) = \nu$. In either case $\mathcal{V}_t$ is the set of admissible controls on $[t, 1]$.

When $c(x, v) = \|v\|_2^2$, this is exactly the Benamou-Brenier formulation of optimal transport

$$\mathcal{W}_2^2(\mu_0, \nu) = \min_{v \in \mathcal{V}_0} \int_0^1 \int \|v_t(x)\|_2^2 d\mu_t(x) dt \quad (22)$$

$$\text{subject to } \frac{d}{dt} \mu_t + \nabla \cdot (v_t \mu_t) = 0 \text{ for all } t \in [0, 1], \quad (23)$$

$$\begin{aligned} \mu_0 &= \mu_0, \\ \mu_1 &= \nu \\ t &\mapsto \mu_t \text{ a.c. and } \mathcal{P}_2(\mathcal{S}). \end{aligned} \quad (24)$$

Discretising the problem as in Müller et al. (2026), we can solve this via a primal-dual approach, including with only sample access to ν .

This forgets about the stationarity completely, but notice that any convex function of v induces a stationary control and thus immediate stationarity of the velocity.

Something that could be interesting is to include in the cost a term such as $\|x\|_1$ to encourage a turnpike behaviour, pushing into a latent space and back out of it. We can also replace the squared norm with $s \|v\|_2^2$ or with $\|v\|_2$ in order to encourage faster convergence to the target distribution from μ_0 .

3. Thinking about mean-field controls

Another way to think about the drifting model optimisation is to treat the empirical data as a finite-particle approximation of a mean-field control problem. In particular, we can view (3) as the mean-field control problem

$$\begin{aligned} J_p(\mu_0) &:= \min_{v \in \mathcal{V}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t c(X_t^v, \mathcal{L}(X_t^v), v_t) \right] \\ \text{subject to } X_{t+1}^v &= X_t^v + v_t(X_t^v) \text{ for all } t \geq 0, \\ X_0^v &\sim \mu_0, \end{aligned} \quad (25)$$

where c is now a function of the law of X_t^v and the velocity field v_t . If we take, for instance, $c(x, \mu, v) = \|v\|_2^2 + \mathcal{W}_2^2(\mu, \nu)$, then the optimal control problem becomes

$$\begin{aligned} J_p(\mu_0) &:= \min_{v \in \mathcal{V}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \|v_t(X_t^v)\|_2^2 + \mathcal{W}_2^2(\mathcal{L}(X_t^v), \nu) \right] \\ \text{subject to } X_{t+1}^v &= X_t^v + v_t(X_t^v, \mathcal{L}(X_t^v)) \text{ for all } t \geq 0, \\ X_0^v &\sim \mu_0, \end{aligned} \quad (26)$$

which is (we can always change the decay scheme of costs to have finite horizon) a penalised version of the Benamou-Brenier transport problem which encourages convergence to the target distribution ν . Of course, the downside of this is that the velocity field will depend on the attraction of all particles to all samples, just as in the original drifting model.

Below is a small generated overview of what we can hope to do in terms of resolution. Basically, everything will work the same way as with non-measure-valued states, but there will be a nastier measure-valued Bellman equation (28) to solve. However, the resulting optimal control is still a feedback control, and there is a rich literature on the resolution using, say JKO-type schemes. The path to samples it to view our algorithm as directly being the numerical scheme on this problem, including a sample collected to represent the two measures μ and ν via propagation of chaos.

Restricting to Markov feedback controls $v_t(\cdot) = \alpha(\cdot, \mu_t)$ with $\mu_t = \mathcal{L}(X_t)$, the induced one-step map is

$$T_\mu^\alpha(x) := x + \alpha(x, \mu), \quad \mu^+ = (T_\mu^\alpha)_\# \mu.$$

The dynamic programming principle on $\mathcal{P}_2(\mathcal{S})$ then takes the form

$$J_p(\mu) = \inf_{\alpha} \left\{ \int_{\mathcal{S}} c(x, \mu, \alpha(x, \mu)) \mu(dx) + \gamma J_p((T_\mu^\alpha)_\# \mu) \right\}. \quad (27)$$

This is the natural discrete-time Bellman equation on the Wasserstein space of probability measures; see for instance [Pham and Wei \(2017\)](#); [Bensoussan et al. \(2013\)](#); [Carmona and Delarue \(2018a\)](#). In particular, the optimal feedback should be characterised by minimisers of the right-hand side of (27), exactly as in the finite-dimensional case, but now after lifting the state variable from x to the whole law μ .

Assuming that J_p is differentiable in the sense of Lions¹, one may define the decoupling field

$$U(\mu, x) := \frac{\delta J_p}{\delta \mu}(\mu)(x),$$

and then formally differentiate (27) with respect to μ . If $\alpha^*(\cdot, \mu)$ is an optimal feedback and $T_\mu^*(x) := x + \alpha^*(x, \mu)$ with $\mu^+ = (T_\mu^*)_\# \mu$, this leads to the discrete-time master equation

$$\begin{aligned} U(\mu, x) = & c(x, \mu, \alpha^*(x, \mu)) + \gamma U(\mu^+, T_\mu^*(x)) + \int_S \frac{\delta c}{\delta \mu}(y, \mu, \alpha^*(y, \mu))(x) \mu(dy) \\ & + \gamma \int_S D_x U(\mu^+, T_\mu^*(y)) \cdot \frac{\delta T_\mu^*}{\delta \mu}(y)(x) \mu(dy), \end{aligned} \quad (28)$$

again at a purely formal level. Equation (28) is the discrete analogue of the continuous-time master equation from mean-field game and mean-field type control theory, and should be understood as a fixed-point equation for the Lions derivative of the value functional; compare with [Cardaliaguet et al. \(2019\)](#); [Carmona and Delarue \(2018b\)](#). It may be possible to formulate this more cleanly by first passing to a finite-horizon problem and then sending the horizon to infinity.

As for solving the problem, the quadratic term $\|v\|_2^2$ by itself does not make the optimisation a simple least-squares problem: the nonlinearity is hidden in the pushforward constraint $\mu^+ = (\text{id} + \alpha(\cdot, \mu))_\# \mu$ and in the dependence of the running cost on the evolving law. Thus a direct calculus-of-variations approach is plausible only in special regimes where one can relax the control problem to a convex optimisation over curves of measures or transport plans, as in Benamou–Brenier or JKO-type formulations; see [Santambrogio \(2015\)](#); [Jordan et al. \(1998\)](#); [Ambrosio et al. \(2005\)](#). In the general mean-field control setting one typically needs either the dynamic programming/master equation route or the stochastic/variational maximum principle developed in [Bensoussan et al. \(2013\)](#); [Carmona and Delarue \(2018a,b\)](#).

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1. See e.g. [Bao et al. \(2021\)](#).

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